

Multimodal RAG: Project Documentation

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Overview

This project tests whether a cross-modal verification layer makes multimodal retrieval-augmented generation (RAG) more accurate and less prone to hallucination on the MMMU benchmark. Four systems are compared on the same pilot data, starting with three naive baselines and ending with the proposed method.

So far the work covers environment setup, a full profile of the MMMU validation split (900 questions across 30 subjects), and the evaluation design. No model has been run yet; baseline numbers come next.

Research question: Does verifying retrieved text, tables and figures against each other before generation improve factual correctness and faithfulness compared with passing unfiltered multimodal evidence to the model and How effectively can independent, evidence-grounded, claim-level verification reduce factual hallucinations in multimodal retrieval-augmented question answering?

Research Hypotheses

Adding evidence-grounded cross-model verification to multimodal RAG reduces factual hallucinations relative to standard multimodal RAG.

Claim-level verification identifies unsupported or contradicted statements more reliably than judging an answer only as a whole.

Evidence-based revision and abstention reduce unsupported claims, but may reduce answer coverage or increase latency.

Dataset: MMMU

MMMU (Massive Multi-discipline Multimodal Understanding, [MMMU/MMMU](#) on Hugging Face) is the benchmark for all four systems. The notebook uses the validation split: 30 subjects with 30 questions each, 900 questions in total.

Subjects (configs): Accounting, Agriculture, Architecture_and_Engineering, Art, Art_Theory, Basic_Medical_Science, Biology, Chemistry, Clinical_Medicine, Computer_Science, Design, Diagnostics_and_Laboratory_Medicine, Economics, Electronics, Energy_and_Power, Finance,

Geography, History, Literature, Manage, Marketing, Materials, Math, Mechanical_Engineering, Music, Pharmacy, Physics, Psychology, Public_Health, Sociology.

Experimental design

Four systems are run on the same MMMU questions, so any difference in score comes from how evidence is retrieved and checked, not from the data.

#	System	Retrieval	Evidence passed to the model	Verification
1	No RAG	None	Question + image(s) only	None
2	Text-only RAG	DPR or BM25	Retrieved text, no image or table context	None
3	Standard multimodal RAG	Text + images/tables	Retrieved text, images and tables in the context window	None
4	Cross-modal verification RAG (proposed)	Text + images/tables	Same as 3, after filtering	Cross-modal check

Baseline 1 uses direct prompting of a multimodal LLM, for example GPT-4o-mini or an open-weight model such as LLaVA or Qwen2-VL.

Evaluation Metrics

Metric	Definition	Notes
QA accuracy / exact match	Prediction matches reference under a documented normalization rule	Primary for multiple-choice QA
Hallucination rate	Fraction of factual claims unsupported or contradicted by evidence	Define annotation protocol first

Verification precision / recall / F1	Verifier decisions compared with human-reviewed claim labels	Requires a labeled claim subset
Evidence faithfulness	Whether answer claims are supported by retrieved evidence	Plausibility is not the same as support
Coverage / abstention rate	Fraction answered versus declined for insufficient evidence	Report alongside accuracy
Latency / cost	Time and API cost per example or condition	Record settings and hardware

Pipelines

No-RAG baseline: MMMU sample (question + image + reference answer) → baseline multimodal model → generated answer → evaluation (accuracy / EM / F1, and hallucination / faithfulness).

RAG variants: MMMU sample → retrieval → top-k evidence → multimodal LLM → RAG answer → the same evaluation. The proposed method adds a cross-modal grounding step between retrieval and the LLM.

The order of work is fixed: record baseline numbers on the pilot dataset first, then build and integrate the proposed contribution.

5. Experimental Conditions

Condition	Pipeline	Purpose
A — Direct VLM	Image + question → generator → answer	Baseline without retrieval or verification
B — Multimodal RAG	Question/image → retrieval → generator → answer	Measure the effect of retrieval
C — RAG + verification	RAG answer → independent evidence verifier	Measure verification performance
D — RAG + verification + revision	Verify → revise or abstain when needed	Measure end-to-end impact and coverage trade-off

Environment setup

The notebook runs on Python 3.13 in a hosted notebook (CPU build of PyTorch) with the libraries below, installed via `pip install -U datasets huggingface_hub transformers accelerate pillow pandas`.

Library	Version	Role
datasets	5.1.0	Loads MMMU and other benchmarks
huggingface_hub	1.33.0	Authentication and artifact sharing
transformers	5.19.0	Open-weight multimodal LLMs and embedding models
accelerate	1.15.0	Efficient model execution
pillow	12.3.0	Image handling for the visual encoders
pandas	3.0.6	Result tables and metric computation

Authentication uses `huggingface_hub.login()` with the device-code flow, and `whoami()` confirmed the session. The pip step also pulled in a newer `huggingface_hub` (2.1.1) than the one imported afterwards (1.33.0), so restart the runtime before running anything that depends on the exact version.

The notebook's CPU-only PyTorch means the open-weight multimodal models (LLaVA, Qwen2-VL) will be slow; a GPU runtime or a hosted API such as GPT-4o-mini is the practical choice for the baseline runs.

Images are spread across seven columns rather than one, so any pipeline has to gather every non-empty slot and keep the question's `<image n>` references aligned with them.

Example

In `validation_Accounting_1` the question asks for the per-unit manufacturing overhead at 30,000 units, and the cost table is an image (733 x 237, RGBA). The options are \$6, \$7, \$8 and \$9, and the answer is B. A text-only system cannot answer it, because the numbers exist only in the picture.

A multi-image case is [validation_Pharmacy_18](#), which has three diagram images (Graph 1 to 3) and asks which one shows transdermal absorption by active transport; the answer is A.

Findings from dataset profiling

95.2% of the 900 validation questions use a single image (857), and 4.8% use two to five images (43). Multi-image questions are rare but are the hardest test for cross-modal grounding, so they deserve their own slice in the results.

Images per question	Questions
1	857
2	24
3	5
4	8
5	6

Difficulty. 424 questions are Medium, 295 Easy and 181 Hard. Of the 43 multi-image questions, 12 are Easy, 18 Medium and 13 Hard, so they span all three levels and image count is not a proxy for difficulty.

Where multi-image questions sit. Art_Theory, Music and Chemistry have 5 each, Psychology 4, Materials and Pharmacy 3 each; 19 of the 30 subjects have at least one, and the other 11 (including Accounting, Finance, Physics and Marketing) have none.

Image types. The Accounting samples inspected are almost all tables, while multi-image questions lean toward diagrams, chemical structures, sheet music and plots. A correct count needs the parsing fix below.

Issues to fix before running models

- **img_type counts are wrong.** `img_type` is stored as a string that looks like a list, so the counting loop tallied individual characters (the output shows letters and brackets). Parse it with `ast.literal_eval` before counting.
- **options has the same string format**, so it also needs parsing before it is shown to a model.
- **Options that are images.** In questions such as `validation_Biology_29` and `validation_Chemistry_26` the choices are `<image n>` placeholders. The prompt has to send those images alongside the text, labelled by option letter.
- **Irregular samples.** `validation_Manage_20` has an empty options list and the answer `2960`, so it is open-ended and needs numeric matching, not letter matching. `validation_Agriculture_26` includes an "I don't know" option.
- **Sparse explanations.** The explanation field is empty for most samples inspected, so it cannot serve as gold reasoning for faithfulness scoring.

Next steps

The notebook saved the metadata table (900 rows: config, id, question, options, answer, img_type, image_count, difficulty) as `mmmu_validation_metadata.csv` and `mmmu_validation_metadata.pkl`. It also defines `inspect_mmmu_sample(config, sample_id)`, a reusable helper that prints a sample's fields and displays its images, and three research subsets: single-image (857), multi-image (43) and all (900).

Next steps, in order:

- Fix `img_type` and `options` parsing, then recompute the image-type counts
- Choose the pilot subset (for example, a few subjects plus all multi-image questions) and the model for Baseline 1
- Write the prompt builder that interleaves question text with every non-empty image slot
- Implement answer extraction and the accuracy / EM / F1 evaluator
- Run Baseline 1 (no RAG) and record accuracy and hallucination rate
- Build the retrieval index and run Baselines 2 and 3, adding retrieval recall and precision
- Implement cross-modal verification (Baseline 4) and rerun the same evaluation